LETTER OF RECOMMENDATION

To Whom It May Concern,

I am pleased to recommend Mr. Nithun Sundarrajan, who completed a Machine Learning Research Internship with Apicore Pharmaceuticals Pvt. Ltd. and the broader RK Pharma Group. His work was brought to my attention on several occasions during his internship, and I am writing this recommendation personally in my capacity as President and Chief Executive Officer of Apicore.

Since 2003, Apicore has served the global pharmaceutical industry as a process research and development and active pharmaceutical ingredient manufacturing partner, with facilities in Vadodara, Gujarat, and Visakhapatnam, Andhra Pradesh. As part of the RK Pharma Group, a vertically integrated pharmaceutical organization with sterile injectable manufacturing operations in the United States, we work in a highly regulated environment where scientific processes, analytical measurements, computerized systems, and documentation are subject to stringent quality and regulatory requirements.

I am a chemist by training and have more than two decades of experience in the pharmaceutical industry. I therefore use the term *rigorous* carefully when evaluating technical work. Nithun's work demonstrated genuine scientific rigor.

His internship focused on identifying areas in which machine learning could provide scientifically meaningful value across API process development, analytical chemistry, and pharmaceutical manufacturing quality operations. This work was strategically relevant as the pharmaceutical industry continues to adopt Quality by Design, Process Analytical Technology, risk-based quality management, and data-driven process understanding in alignment with principles described in ICH Q8 and ICH Q9.

A particularly notable aspect of Nithun's approach was his recognition that pharmaceutical development data are not typically generated for machine-learning purposes. Such data often involve limited sample sizes, complex experimental dependencies, inconsistent formats, and measurements distributed across development and operational systems. Before developing any model, he invested considerable effort in understanding how the data were generated, recorded, reviewed, and used by scientists. This disciplined approach distinguished him from many technically capable researchers.

One of his principal contributions involved the predictive modelling of API process-development data. He organized historical reaction-development records containing variables such as temperature, reaction time, solvent selection, reagent ratios, pH, and intermediate analytical results into structured datasets suitable for predicting yield, assay, and impurity levels. Recognizing the relatively small number of experiments compared with the dimensionality of the process space, he appropriately prioritized data-efficient and interpretable methods. He compared regularized regression, random forests, and gradient-boosted decision trees using cross-validation strategies that preserved the grouping of related experiments. Importantly, he understood that conventional random data splitting could result in data leakage and misleadingly optimistic performance estimates.

His evaluation extended beyond reporting aggregate accuracy. He examined where the models were extrapolating beyond the range of available training data, assessed the stability of model performance, and reviewed whether influential process variables were scientifically and mechanistically plausible to experienced chemists. This combination of statistical discipline and scientific interpretation is particularly important in pharmaceutical process development.

Nithun also evaluated anomaly-detection approaches for analytical and manufacturing data. Because confirmed examples of process failures and abnormal events were limited, he correctly emphasized unsupervised and semi-supervised methods. He assessed principal component analysis, isolation-based algorithms, one-class models, and statistical process-monitoring techniques for identifying atypical impurity profiles and potential early indicators of abnormal operating conditions. He consistently treated anomaly scores as tools for prioritizing expert scientific review rather than as a basis for autonomous quality or manufacturing decisions.

In addition, he completed a feasibility assessment of transfer learning and image-based anomaly detection for visual-inspection and packaging applications. His assessment carefully considered the challenges associated with rare-defect detection, class imbalance, false-rejection rates, inspection sensitivity, data availability, and the substantial difference between promising laboratory results and a method suitable for validated use on a commercial manufacturing line.

Another important contribution was the development of an evidence-grounded document-intelligence prototype using retrieval-augmented generation. The prototype operated over a controlled collection of standard operating procedures, technical reports, and investigation summaries and was designed to reduce the time scientists spend locating relevant technical information. Nithun separated the document-retrieval process from the generation process, ensured that responses were supported by source references, and proposed appropriate safeguards against hallucination, unauthorized access, and the use of obsolete document versions.

His recommendations included document-version filtering, metadata-based access controls, clear source attribution, controlled indexing, and mandatory verification against approved records before any information could be relied upon for regulated activity. In an industry where data integrity, document control, and traceability are fundamental, his decision to prioritize verifiability over conversational fluency reflected sound professional judgement.

Nithun's final deliverables included a structured assessment of our available data landscape, reproducible modelling notebooks, benchmark results, documented assumptions and limitations, and a prioritized technical roadmap for the responsible adoption of machine learning. These outputs continue to inform our broader thinking regarding digital and analytical capabilities.

From a technical perspective, he demonstrated strong proficiency in Python, pandas, scikit-learn, PyTorch, and SQL. His code was organized, modular, and reproducible, and his experimental records were maintained with considerable attention to detail.

What impressed me most was his scientific temperament. He became productive within a highly specialized and regulated domain in a remarkably short period. He did not overstate the significance of his results and consistently distinguished between an encouraging research observation, a technically feasible prototype, and a system suitable for operational or validated use.

He approached machine learning as a decision-support capability for scientists and subject-matter experts rather than as a replacement for scientific judgement. This maturity of perspective is uncommon at his career stage. He also communicated effectively with research scientists, analytical chemists, engineers, quality professionals, and senior leadership, adapting the level of technical detail appropriately for each audience.

Nithun combines genuine research ability with the practical engineering skills required to develop systems under real-world scientific, operational, and regulatory constraints. This combination is rare at any stage of a career and particularly impressive in an undergraduate researcher.

I recommend him enthusiastically and without reservation for research and professional opportunities in machine learning, pharmaceutical data science, scientific artificial intelligence, intelligent manufacturing, and related fields. He would be welcome to work with Apicore again in the future.

Should any additional information be required, my office may be contacted directly.

Sincerely,

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